

Deterministic Optimization

Linear Optimization Modeling
Handling Nonlinearity

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Radiation Therapy using LP

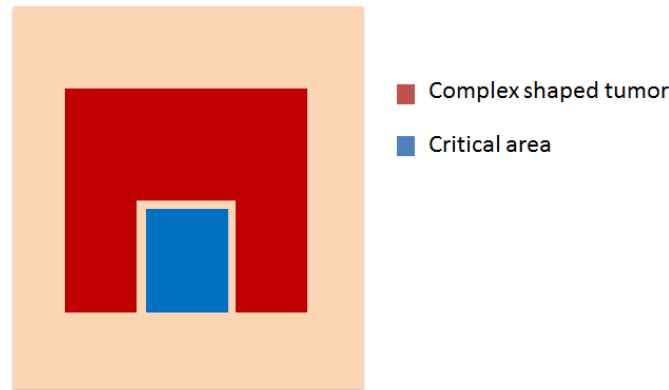
Modeling using Linear Programs

Learning Objectives for this Lesson

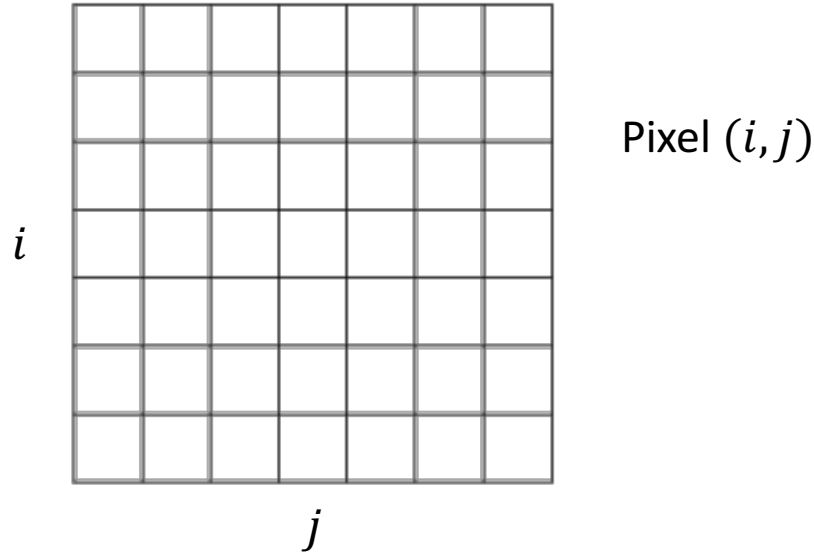
- Create an idealistic LP model for Radiation Therapy Planning
- Create a realistic nonlinear optimization model
- Reformulate as LP

Radiation Therapy Planning Problem

- For a given tumor and given critical areas
- For a given set of possible beamlet origins and angles (we assume angles are fixed, in reality, angles can be optimized also.)
- Determine the weight of each beamlet s.t.
 - Dosage over the tumor area will be at least a target level γ_L
 - Dosage over the critical area will be at most a target level γ_U



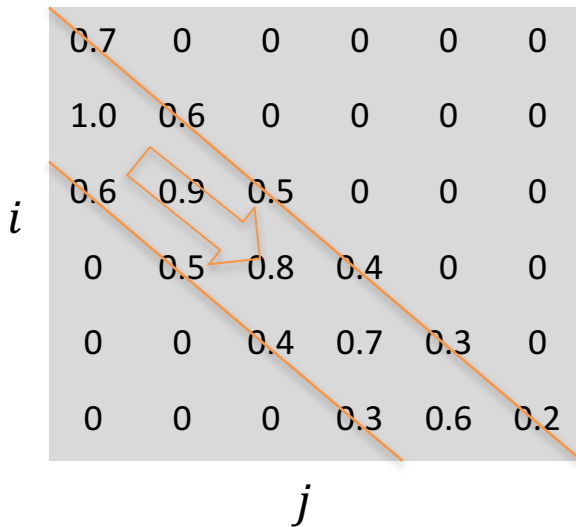
Discretize Radiation Area



Partition radiation area into 2D pixels or 3D voxels

Beamlet Data

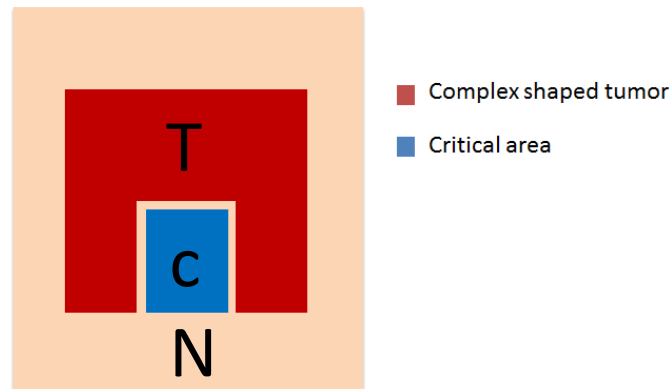
- Beamlet data specifies the unit dose delivery quantity for each beamlet, i.e. the dose amount given a unit radiation from a beamlet
- D_{ij}^p : Dose delivered to pixel (i, j) by beamlet p
- D^p : Dose matrix with entry D_{ij}^p



(Ideal) Linear Optimization Model

- Decision variable:
 - $w = (w_1, \dots, w_N) \in R_+^N$
 - w_p : weight of beamlet p
- Objective: minimize total dose

$$\begin{aligned} \min \quad & \sum_{(i,j) \in S} D_{ij} \\ \text{s.t.} \quad & D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S \\ & w \geq 0 \\ & D_{ij} \geq D_{ij}^L \quad \forall (i,j) \in T \\ & D_{ij} \leq D_{ij}^U \quad \forall (i,j) \in C \end{aligned}$$



T : = set of pixels of tumor area
 C : = set of pixels of critical area
 N : = set of pixels of normal area
 S : = $T \cup C \cup N$

Unfortunately, this model is typically infeasible: i.e. cannot deliver dose to tumor without some harm to critical areas.

Realistic Nonlinear Optimization Model

- We can set a targeted level dose for each pixel Target_{ij}
- Require the delivered dose is as close as possible to Target_{ij}
- Again, we need a metric to measure distance between D_{ij} and Target_{ij}

- Use l_1 -metric: total abs difference

- $$\begin{aligned} \min \quad & \sum_{(i,j) \in S} |D_{ij} - \text{Target}_{ij}| \\ \text{s.t.} \quad & D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S \\ & w \geq 0 \end{aligned}$$

- Use l_∞ -metric: max abs difference

- $$\begin{aligned} \min \quad & \max_{(i,j) \in S} |D_{ij} - \text{Target}_{ij}| \\ \text{s.t.} \quad & D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S \\ & w \geq 0 \end{aligned}$$

LP Reformulations

- Use l_1 -metric: total abs difference
- $\min \sum_{(i,j) \in S} |D_{ij} - \text{Target}_{ij}|$
s.t. $D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S$
 $w \geq 0$
- Use l_∞ -metric: max abs difference
- $\min \max_{(i,j) \in S} |D_{ij} - \text{Target}_{ij}|$
s.t. $D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S$
 $w \geq 0$

$$\begin{aligned} \min \quad & \sum_{(i,j) \in S} z_{ij} \\ \text{s.t.} \quad & -z_{ij} \leq D_{ij} - \text{Target}_{ij} \leq z_{ij}, \forall (i,j) \in S \\ & D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S \\ & w \geq 0 \end{aligned}$$

$$\begin{aligned} \min \quad & z \\ \text{s.t.} \quad & -z \leq D_{ij} - \text{Target}_{ij} \leq z, \forall (i,j) \in S \\ & D_{ij} = \sum_{p=1}^N D_{ij}^p w_p \quad \forall (i,j) \in S \\ & w \geq 0 \end{aligned}$$

Summary for this lesson

- Create an idealistic LP model for Radiation Therapy Planning
- Create a realistic nonlinear optimization model using convex PWL functions
- Reformulate as LP
- Conclude discussion on LP modeling

Summary for LP Modeling

- **Learned three types of network flow optimization models**
 - Transportation problem
 - Max-flow problem/Min-cut problem
 - Shortest path problem
- **Discovered how LP models are used to operate electric markets**
 - Power network flow model
 - Locational marginal prices
- **Formulated stochastic programming**
 - Two-stage stochastic program
- **Utilized convex PWL function**
 - Robust regression
 - Radiation therapy